

Statistical Models

Assignment 1

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Introduction

This assignment concerns maximum likelihood and asymptotic variance, and is motivated by a real-world problem involving *Salmonella enterica* in eggs.

Eggs are thought to be infected with a bacterium *Salmonella enterica* so that the number of organisms, Y , in each egg has a Poisson distribution with mean μ . The value of Y cannot be observed directly, but after a period it becomes certain whether the egg is infected ($Y > 0$) or not ($Y = 0$). Out of m such eggs, r are found to be infected.

Question (a)

We wish to show that the maximum likelihood estimator $\hat{\mu}$ of μ is

$$\hat{\mu} = \log \left(\frac{m}{m-r} \right).$$

We consider the event $P(Y_i > 0)$, where $Y_i \sim \text{Poisson}(\mu)$ is the number of organisms in the i -th egg, and we may state the number of infected eggs as.

$$R = \sum_{i=1}^m \mathbf{1}(Y_i > 0).$$

The probability $P(Y_i > 0) = 1 - P(Y_i \leq 0)$ can be calculated from the CDF of the Poisson distribution

$$1 - e^{-\mu} \sum_{j=0}^0 \frac{\mu^j}{j!} = 1 - e^{-\mu}.$$

So R is the sum of m independent Bernoulli random variables with success probability $p(\mu) = 1 - e^{-\mu}$. Thus R has a binomial distribution with parameters m and p . The MLE for a binomial distribution is the fractions of succesful trials to the total number of trials. Note that $p(\mu)$ is a strictly monotone function of μ , as such we can find $\hat{\mu} = f^{-1}(\hat{p})$ and the inverse is guaranted to exist.

$$\hat{p} = \frac{r}{m} \iff 1 - e^{-\hat{\mu}} = \frac{r}{m} \iff e^{-\hat{\mu}} = 1 - \frac{r}{m} \iff e^{-\hat{\mu}} = \frac{m-r}{m} \iff \hat{\mu} = \log\left(\frac{m}{m-r}\right) \quad (1)$$

□

Question (b)

Since R is a sample with just one realization the likelihood function is given by the probability mass function of the binomial distribution, let $c = \binom{m}{r}$, then the likelihood function is

$$L(p) = P(R = r) = cp^r(1-p)^{m-r}.$$

with log-likelihood function

$$\ell(p) = \log L(p) = \log c + r \log p + (m-r) \log(1-p).$$

If we substitute $p = 1 - e^{-\mu}$ we get

$$\ell(\mu) = \log c + r \log(1 - e^{-\mu}) + (m - r) \log(e^{-\mu}) = \log c + r \log(1 - e^{-\mu}) - (m - r)\mu.$$

The score function is given by the derivative of the log-likelihood function with respect to μ ,

$$S(\mu) = \frac{d\ell}{d\mu} = \frac{re^{-\mu}}{1 - e^{-\mu}} - (m - r).$$

We then find the second derivative of the log-likelihood function with respect to μ ,

$$\frac{d^2\ell}{d\mu^2} = \frac{dS(\mu)}{d\mu} = \frac{-re^{-\mu}(1 - e^{-\mu}) - re^{-\mu}e^{-\mu}}{(1 - e^{-\mu})^2} = \frac{-re^{-\mu}}{(1 - e^{-\mu})^2}.$$

The Fisher information is given by the negative expected value of the second derivative of the log-likelihood function with respect to μ ,

$$I(\mu) = E\left[-\frac{d^2\ell(\mu)}{d\mu^2}\right] = E\left[\frac{Re^{-\mu}}{(1 - e^{-\mu})^2}\right] = \frac{e^{-\mu}}{(1 - e^{-\mu})^2} E[R] = \frac{e^{-\mu}}{(1 - e^{-\mu})^2} mp$$

Since $p = 1 - e^{-\mu}$ we have

$$I(\mu) = \frac{e^{-\mu}}{(1 - e^{-\mu})^2} m(1 - e^{-\mu}) = \frac{me^{-\mu}}{1 - e^{-\mu}}.$$

Since our entire sample consists of just one realization of R , the asymptotic variance of $\hat{\mu}$ is given by the inverse of the Fisher information,

$$\text{Var}(\hat{\mu}) = I(\mu)^{-1} = \frac{1 - e^{-\mu}}{me^{-\mu}} = \frac{e^{\mu} - 1}{m}$$

□

Question (c)

We run simulations to estimate the mean and variance of $\hat{\mu}$, and to plot the empirical CDF of $\hat{\mu}$. We use $Nsim = 10000$ simulations, with $m = 20$ eggs and $\mu = 2$. I find the estimated mean and variance:

$$\hat{\mu} = 2.08, \quad \text{Var}(\hat{\mu}) \approx 0.299 \quad (2)$$

Note that since we have a large number of simulations, we will sometimes hit a sample where all eggs are infected, i.e. $r = m$. In this case we have a division by zero error when calculating $\hat{\mu}$, and we simply skip that sample.

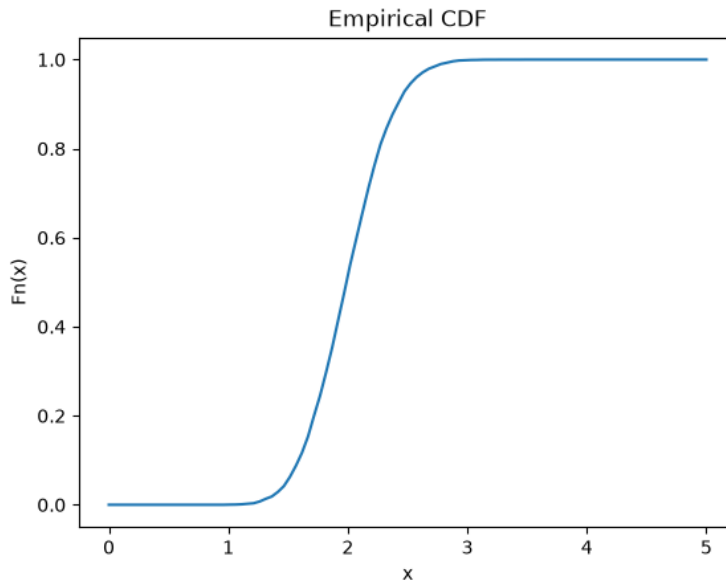


Figure 1: Empirical CDF of $\hat{\mu}$.

The code for the simulations is shown below.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 np.random.seed(1)
4 Nsim = 10000
5 m = 20
6 mu = 2
7 xrange = np.linspace(0,5,100)
8
9 means = []
10
11 for i in range(Nsim):
12     # Unobservable Poission
13     y = np.random.poisson(mu, m)
14     # How many of the eggs have bacteria?
15     r = sum([1 if yi > 0 else 0 for yi in y])
16     # Division by zero error!
17     try:
18         mu_hat = np.log(m / (m-r))
19     except ZeroDivisionError:
20         continue
21     means.append(mu_hat)
22
23
24 Femp = lambda means: [np.sum(means < x) / len(means) for x in xrange]
25 print("Estimated mean: ", np.mean(means))
26 print("Estimated variance: ", np.var(means))
27 theoretical_var = (np.exp(mu) - 1) / m
28 print("Theoretical variance: ", theoretical_var)
29
30 plt.plot(xrange, Femp(means))
31 plt.xlabel("x")
32 plt.ylabel("Fn(x)")
33 plt.title("Empirical CDF")
34 plt.savefig("CDF2.png")
35 plt.close()

```

The empirical CDF of $\hat{\mu}$ is shown in Figure 1, and has a striking resemblance to the theoretical CDF for a normal distribution.

Question (d)

We now redo the simulations with $m = 200$ eggs, we get the CDF in Figure 2. The estimated mean and variance are

$$\hat{\mu} = 2.016, \quad \text{Var}(\hat{\mu}) \approx 0.034 \quad (3)$$

The theoretical variance is given by

$$\text{Var}(\hat{\mu}) = \frac{e^{\mu} - 1}{m} = \frac{e^2 - 1}{200} \approx 0.0319 \quad (4)$$

We see that the CDF steepens around the true value of $\mu = 2$, and the estimated variance is closer to the theoretical variance. This is consistent with the asymptotic normality of $\hat{\mu}$, as we have increased the sample size m .

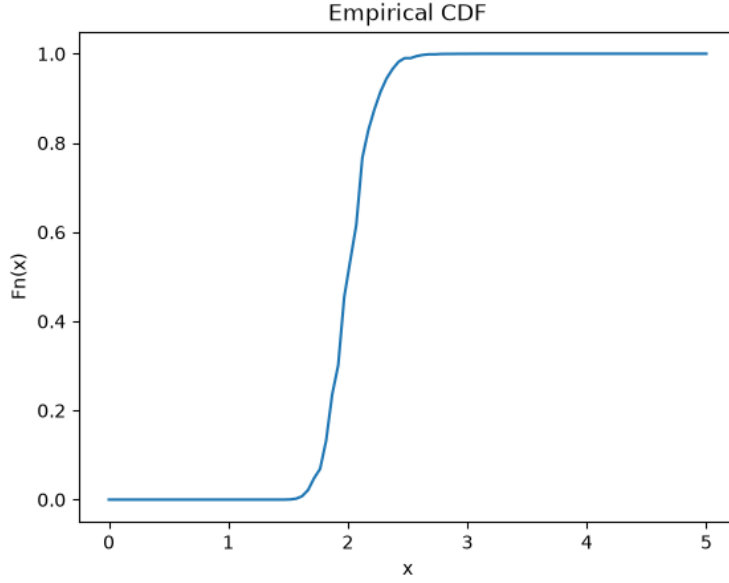


Figure 2: Empirical CDF of $\hat{\mu}$ with $m = 200$.

Question (e)

We now address division by zero errors. Due to the finite sample size and non-zero probability of having all eggs infected, we will sometimes hit a sample where $r = m$. According to our model this is a Binomial experiment with only successes, thus the probability of this event is given by

$$P(R = m) = p^m = (1 - e^{-\mu})^m$$

Since $e^x > 0$ for all $x \in \mathbb{R}$, we have that $(1 - e^{-\mu}) < 1$ for all $\mu > 0$. Thus, if we increase the sample size m , the probability of division by zero errors decreases exponentially, so

$$P(R = m) = (1 - e^{-\mu})^m \rightarrow 0 \quad \text{as } m \rightarrow \infty. \quad (5)$$

Therefore, the probability of division by zero errors approaches zero as the sample size m increases - which is consistent with the asymptotic normality of $\hat{\mu}$. However, it highlights issues with finite sample approximations. We illustrate this in the QQ-plots in Figure 3. We see that the normal approximation of $\hat{\mu}$ improves as m increases. The code for the QQ-plots is shown below.

```

1 from scipy.stats import probplot
2 plt.figure(figsize=(12, 10))
3
4 for j, m in enumerate([20, 50, 100, 500]):
5     mu_hats = []
6     for i in range(Nsim):
7         y = np.random.poisson(mu, m)
8         r = sum([1 if yi > 0 else 0 for yi in y])
9         try:
10             mu_hat = np.log(m / (m - r))
11         except ZeroDivisionError:
12             continue
13         mu_hats.append(mu_hat)
14
15     plt.subplot(2, 2, j + 1)
16     probplot(mu_hats, dist="norm", plot=plt)
17     plt.title(f"Normal Q-Q plot, m = {m}")
18
19 plt.tight_layout()
20 plt.savefig("QQ.png")
21 plt.show()

```

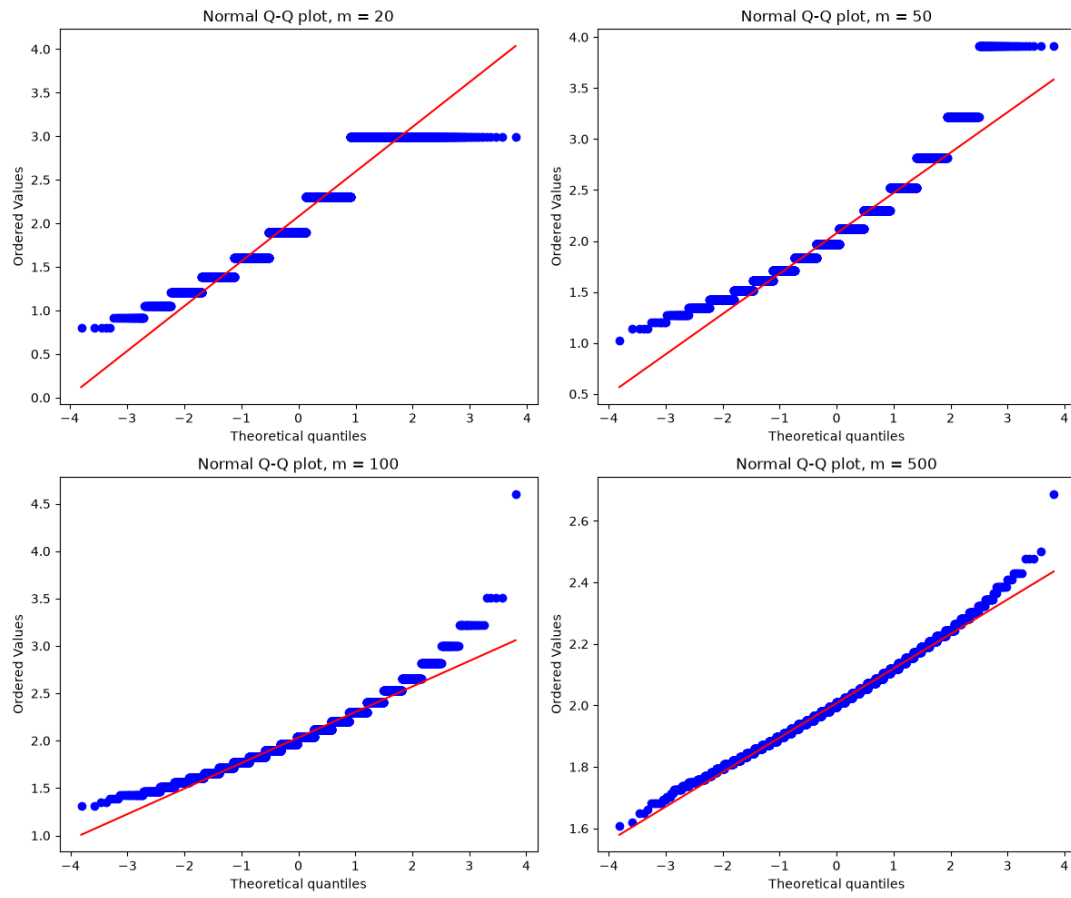


Figure 3: QQ-plots at $m = 20, 50, 100, 500$